

**FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION,
ISLAMABAD**

Annual Examination 2026

Biology — Class IX

Syllabus: Ch 1 — The Science of Biology | Ch 2 — Biodiversity | Ch 3 — The Cell | Ch 4 — Cell Cycle | Ch 5 — Tissues, Organs & Organ Systems | Ch 6 — Molecular Biology | Ch 7 — Metabolism (Sec 7.1-7.6)

Total Marks: 65

Time Allowed: 2 Hours 30 Minutes

Version: 2

General Instructions:

1. This paper has three sections: Section A (MCQs), Section B (Short Questions), and Section C (Long Questions).
2. Attempt Section A on the question paper by filling the bubble next to the correct option.
3. In Section B, attempt any 11 questions. For each, choose either the main question OR its alternative.
4. In Section C, attempt all 4 long questions. For each, choose either the main question OR its alternative.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	The branch of Biology that deals with the study of internal body structures is called:	Physiology	Morphology	Anatomy	Histology	○○○○
2	In the five-kingdom classification, the kingdom that includes single-celled eukaryotes is:	Monera	Fungi	Protista	Plantae	○○○○
3	Viruses are said to be "on the boundary of life" because they:	have a cell wall	can be crystallized but also reproduce	perform respiration	belong to Domain Bacteria	○○○○
4	The cell wall of fungi is composed of:	Cellulose	Pectin	Chitin	Peptidoglycan	○○○○
5	During which phase of the cell cycle does DNA replication occur?	G1 phase	S phase	G2 phase	M phase	○○○○
6	Which organ system includes the kidneys and bladder and is responsible for removing nitrogenous waste?	Endocrine system	Lymphatic system	Urinary system	Digestive system	○○○○
7	Palisade mesophyll cells in a leaf are tightly packed and contain many chloroplasts mainly to:	allow gas diffusion	maximise light absorption	provide mechanical support	transport water upward	○○○○
8	The generalized formula of carbohydrates is:	$C_xH_{2y}O_y$	$C_x(H_2O)_y$	CH_2O	$C_6H_{12}O_6$	○○○○
9	The process by which DNA sequence is copied into mRNA is called:	Translation	Replication	Transcription	Transduction	○○○○
10	Which type of inhibition can be reversed by increasing the concentration of substrate?	Non-competitive	Allosteric	Competitive	Permanent	○○○○
11	The non-competitive inhibitor binds to the enzyme at a site OTHER than the active site called the:	Prosthetic site	Allosteric site	Coenzyme site	Substrate site	○○○○
12	Catabolism is best described as reactions that:	build complex molecules from simple ones	require light energy only	break complex molecules into simpler ones	synthesize proteins from amino acids	○○○○

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt 11 questions. Choose either the main question OR its alternative.

Part	Question	Marks	OR	Alternative Question	Marks
2(i)	What is the difference between biology and biochemistry? Name any FOUR sub-fields of biology with a one-line description of each.	3	OR	Describe THREE careers in biology. For each career, name the degree required and the type of work involved.	3
2(ii)	What is a hypothesis? How is it different from a theory? Give one example of each from the malaria case study.	3	OR	Explain the role of Alphonse Laveran (1882) and Ronald Ross (1897) in discovering the cause and transmission of malaria.	3
2(iii)	Compare the two-kingdom and five-kingdom systems of classification. Who proposed each, and what three kingdoms were added in the five-kingdom system?	3	OR	Explain the domain Archaea. Name THREE types of archaeobacteria, their habitats, and why they differ from bacteria.	3
2(iv)	Describe the structure and function of the Golgi apparatus. Who discovered it? Name TWO products it forms.	3	OR	Compare the structure and function of chloroplasts and mitochondria. Name ONE specific reaction that occurs in each organelle.	3
2(v)	Describe the events during Metaphase and Anaphase of mitosis. What is the role of the spindle apparatus in each stage?	3	OR	What is the G0 phase? Name TWO cell types that enter G0. Explain why cancer cells do NOT enter G0.	3
2(vi)	Explain what happens during Prophase I of meiosis. Define synapsis, bivalent, and crossing over.	3	OR	State FOUR significances of meiosis. How does meiosis contribute to genetic variation?	3
2(vii)	Define homeostasis. Describe the homeostatic response using: stimulus, sensor, control centre, effector, and negative feedback.	3	OR	Name and describe the FOUR types of animal tissues. Give one example of each.	3
2(viii)	Explain how the shoot system and root system of a plant differ in structure and function.	3	OR	What are spongy mesophyll cells? Explain their role in gas exchange. How do they differ structurally from palisade mesophyll cells?	3
2(ix)	Distinguish between saturated and unsaturated fatty acids. Give TWO examples of each and state which is solid and which is liquid at room temperature.	3	OR	State FOUR functions of lipids in living organisms. Why does one gram of lipid yield more energy than one gram of carbohydrate?	3
2(x)	What are polysaccharides? Name THREE polysaccharides and state the role of each in living organisms.	3	OR	Explain the structure of a nucleotide. Name the TWO types of pentose sugars in nucleic acids and state which is in DNA and which in RNA.	3
2(xi)	Explain how enzyme activity is affected by changes in pH. Give ONE example each of an enzyme that works best in acidic and in alkaline conditions.	3	OR	Describe THREE types of cofactors: activator, prosthetic group, and coenzyme. Give one example of each.	3

SECTION C – Long Questions (4 × 5 = 20 Marks)

Part	Question	Marks	OR	Alternative Question	Marks
3(i)	Explain the biological method in science. Describe all SIX steps in sequence with an example from the malaria case study. Also state the difference between a scientific law and a scientific theory.	5	OR	Explain the three-domain system proposed by Carl Woese: the three domains, their key characteristics, and the four kingdoms of Domain Eukarya.	5
3(ii)	Describe the structure and functions of the nucleus. Include nuclear envelope, nuclear pores, chromatin, nucleolus, and explain how chromatin forms chromosomes. Also explain the role of histones and define gene and genome.	5	OR	Describe the complete process of meiosis I and meiosis II. Name and describe all stages of each division. Explain how meiosis differs from mitosis in number of divisions, chromosome number in daughter cells, and genetic outcome.	5
3(iii)	Explain the structure of DNA (Watson and Crick model). State SEVEN key points of the model. Explain complementary base pairing, hydrogen bonds, and define gene and genome.	5	OR	Classify carbohydrates into monosaccharides, disaccharides, and polysaccharides. Give TWO examples and molecular formula of each type. Explain how sucrose, maltose, and lactose are formed.	5
3(iv)	Describe the characteristics of enzymes in detail. Name and explain any SIX characteristics. Include the concepts of activation energy, cofactors, and metabolic pathway.	5	OR	Explain how substrate concentration and temperature each affect enzyme activity. Define denaturation, optimum temperature, and saturation of active site.	5